

Notes: Eaton, Kortum, and Kramarz (Econometrica, 2011)

An Anatomy of International Trade: Evidence from French Firms

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1 Big picture

- **Question.** What can firm-level export data teach us about the sources of international trade, the role of firm heterogeneity, and the firm-level consequences of trade liberalization?
- **Setting.** The paper studies French manufacturing firms selling in 113 destinations, including France itself. The data combine firm domestic sales, firm exports by destination, and aggregate market-size and trade-share information.
- **Empirical facts.** Larger markets attract more French exporters; normalized entry rises systematically with market size; sales distributions look strikingly similar across destinations; and firms that sell to more destinations or more popular destinations also sell more at home.
- **Model.** The authors build a structural model of monopolistic competition with firm heterogeneity, destination-specific entry shocks, demand shocks, fixed costs of market access, and aggregate gravity-like trade shares.
- **Estimation.** Parameters are estimated by simulated moments using the observed firm-destination entry matrix and sales distributions for French exporters.
- **Main decomposition.** A single dimension of firm heterogeneity—underlying efficiency—explains more than half of cross-firm variation in market entry. Conditional on entry, however, sales variation within a destination is driven much more by destination-specific shocks.
- **Counterfactual.** A 10 percent decline in bilateral trade barriers raises total French firm sales by about \$16 billion and exports by about \$35 billion, but the gains are concentrated in the largest firms.
- **Distributional result.** Trade liberalization expands exports and the number of exporters, but it also reduces domestic sales and pushes many small firms out. The top decile gains; lower deciles lose sales.
- **Takeaway.** Aggregate gains from trade hide large within-country reallocation. Export opportunities benefit the most productive firms disproportionately while exposing lower-productivity firms to stronger import competition at home.

2 Incremental contribution of the paper

2.1 Relative to aggregate gravity and Eaton–Kortum style models

- Earlier aggregate trade models, especially Eaton and Kortum style Ricardian models, explain bilateral trade shares using aggregate productivity dispersion and trade costs.
- Those models discipline country-level trade flows but leave the distribution of individual producers hidden.
- Eaton, Kortum, and Kramarz add firm-level data to the aggregate gravity tradition. They ask whether a structural model can simultaneously rationalize country-level trade shares and the granular distribution of firm entry and sales.

Incremental contribution: The paper turns aggregate trade theory into an empirically disciplined firm-level model. It uses the firm-destination matrix to identify which components of heterogeneity are responsible for entry, sales, and counterfactual gains.

2.2 Relative to Melitz and heterogeneous-firm trade models

- Melitz-style models emphasize firm productivity and selection into exporting.
- A standard implication is that more productive firms export and larger or more productive exporters serve more markets.

- The French data reveal more structure: exporters form a hierarchy across destinations, sales distributions are similar across markets, and export intensity varies systematically with destination popularity.
- A one-dimensional productivity model alone cannot match both entry patterns and sales dispersion conditional on entry.

Incremental contribution: The paper shows that efficiency matters enormously for entry but is insufficient for explaining conditional sales. Destination-specific demand and entry shocks are needed to reconcile the observed sales distributions and export intensity patterns.

2.3 Relative to partial-equilibrium export-entry studies

- Earlier firm-level export studies often modeled export participation as a partial-equilibrium decision of the firm.
- This paper embeds firm choices into a general-equilibrium trade structure with market sizes, price indexes, wages, trade shares, and counterfactual equilibrium changes.
- The same estimated model can match micro moments and generate macro counterfactuals.

Incremental contribution: The paper connects micro export decisions to aggregate trade equilibrium. This is essential because trade liberalization changes both foreign opportunities and domestic competition.

2.4 New counterfactual insight

- A uniform fall in trade barriers is not simply good for all firms.
- The model predicts large sales increases for the most productive exporters, but sales losses for most lower deciles.
- Domestic sales decline because foreign firms become more competitive in France; export opportunities are concentrated among the most efficient French firms.

Incremental contribution: The paper is one of the clearest early structural decompositions of aggregate trade-liberalization effects into firm-level winners and losers.

3 Data and measurement

3.1 Firm-level sales and export data

- The core data are a cross-section of French manufacturing firms.
- For each firm, the authors observe domestic sales in France and exports by destination.
- The destination set contains 113 markets, including France.
- In the baseline data, there are roughly 231,000 French firms and about 34,000 exporters.
- Exports are observed at the firm-destination level, which allows the authors to construct each firm's portfolio of foreign markets.

Interpretation: The data are unusually rich because they reveal both the extensive margin—where each firm sells—and the intensive margin—how much it sells in each market.

3.2 Destination-level variables

- For each destination, the paper uses aggregate market size, French sales, French market share, and the number of French firms entering the market.
- Market size is measured in dollars and is used to study how entry and sales vary with destination size.
- The top foreign destinations for French exporters include Belgium-Luxembourg, Germany, Switzerland, Italy, the United Kingdom, the Netherlands, and the United States.

Interpretation: Destination size and popularity are essential for identifying entry costs and the hierarchy of export participation.

3.3 Entry variables

For a French firm j and destination n , define an entry indicator

$$d_{nj} = 1\{\text{firm } j \text{ sells in destination } n\}. \quad (1)$$

The number of French firms selling in destination n is

$$N_{nF} = \sum_j d_{nj}. \quad (2)$$

Interpretation: The vector $(d_{1j}, \dots, d_{Nj})$ is the firm's market portfolio. Its pattern across destinations reveals the selection role of firm efficiency and market-entry costs.

3.4 Sales variables

Let $X_{nF}(j)$ denote sales of French firm j in destination n , and let $\bar{X}_{nF}$ denote average sales by French firms in destination n .

The paper uses several transformed outcomes:

- firm sales in each destination, normalized by average sales in that destination;
- firm sales in France conditional on entry into foreign markets;
- normalized export intensity, defined as export sales in a market relative to the firm's French sales, normalized by the market average.

Interpretation: Normalization is crucial because destinations differ greatly in size. The key question is whether distributions have similar shapes after adjusting for scale.

4 Empirical regularities

4.1 Entry rises with market size

- Larger destinations attract more French exporters.
- The number of French exporters relative to French market share rises systematically with destination size.
- This pattern is approximately log-linear.

Interpretation: Entry costs cannot be independent of market size in a simple way. Large markets generate more entry than implied by sales shares alone.

4.2 Hierarchy among popular export destinations

- French exporters tend to enter popular destinations in a hierarchical sequence.
- For the top-seven foreign destinations, many firms sell to Belgium only; some sell to Belgium and Germany; fewer sell to Belgium, Germany, and Switzerland; and so forth.
- Under independent entry across destinations, the number of firms following this hierarchy would be much smaller.

Interpretation: The hierarchy suggests that a common firm-level component, such as efficiency, strongly shapes entry across markets.

4.3 Similar sales distributions across destinations

- Sales distributions look remarkably similar across markets of very different size and French participation.
- The rank-sales curves for Belgium-Luxembourg, France, Ireland, and the United States have similar shapes after normalizing by market mean sales.
- The upper tail resembles a Pareto distribution, but the lower tail has curvature.

Interpretation: A model must explain why destination size changes the scale of sales but not much of the distributional shape.

4.4 Sales in France and export participation

- Firms that enter more destinations have higher sales in France.
- Firms that enter more popular foreign destinations also sell more in France.
- The relation between French domestic sales and the number of firms entering a destination is tight and negative in logs: a less popular destination is reached only by larger domestic firms.

Interpretation: Domestic sales reveal latent firm efficiency. More efficient firms are larger at home and can overcome harder foreign market-entry hurdles.

4.5 Export intensity by destination

- Export intensity is small relative to home sales for most destination-firm pairs.
- Normalized export intensity rises with the popularity of the destination.
- Even among exporters, foreign sales are often much smaller than domestic sales.

Interpretation: Export participation does not mean that foreign markets become equally important for the firm. The model must allow partial reach and destination-specific sales shocks.

5 Model and key equations

5.1 Demand and pricing

Firms sell differentiated goods under monopolistic competition. A compact representation of destination demand is

$$X_{ni}(j) = a_n(j) \left(\frac{p_{ni}(j)}{P_n} \right)^{1-\sigma} X_n, \quad (3)$$

where $a_n(j)$ is a firm-destination demand shock, $p_{ni}(j)$ is the price of firm j from source i in destination n , P_n is the destination price index, X_n is total expenditure, and $\sigma > 1$ is the elasticity of substitution.

With constant markups,

$$p_{ni}(j) = m c_{ni}(j), \quad m = \frac{\sigma}{\sigma - 1}. \quad (4)$$

Interpretation: The demand shock affects sales conditional on entry, while costs and fixed costs determine market selection.

5.2 Entry condition

A firm enters destination n when its unit cost is low enough relative to a destination-specific entry hurdle:

$$c_i(j) < \bar{c}_{ni}(\eta_n(j)), \quad (5)$$

where $\eta_n(j)$ is an entry shock. The paper derives the market-specific threshold from profits net of fixed costs.

Interpretation: Efficiency is common across destinations, while entry shocks vary by destination. This structure generates both hierarchical entry and deviations from a strict hierarchy.

5.3 Efficiency transformation

The paper transforms French firm efficiency into a uniform latent variable:

$$u(j) = T_F z_F(j)^{-\theta}. \quad (6)$$

Here $z_F(j)$ is firm efficiency and θ governs dispersion. Lower $u(j)$ means higher efficiency.

Interpretation: This transformation makes the simulation algorithm straightforward: firms draw a latent efficiency rank, and market-specific entry hurdles determine where they sell.

5.4 Destination-specific sales conditional on entry

In the streamlined representation, sales conditional on entry can be written as a function of three components:

$$X_{nF}(j) = \text{scale}_n \times \text{firm efficiency component}_j \times \text{destination-specific shock}_{nj} \times \text{buyer-reach term}_{nj}. \quad (7)$$

The paper's exact equation (31) implies that, after normalizing by destination scale, sales distributions are common across destinations except for destination-specific shocks and buyer reach.

Interpretation: Efficiency affects sales, but conditional sales are not determined by efficiency alone. Demand and reach shocks matter substantially.

5.5 Sales in France conditional on foreign entry

The model links sales in France of firms that enter destination n to the number of French firms entering that destination:

$$X_{FF}(j \mid j \in n) \text{ declines with } N_{nF}. \quad (8)$$

In the model, entering a less popular destination means passing a harder hurdle, so the firms observed in that destination tend to have higher home sales.

Interpretation: This prediction explains Figure 3: the fewer French firms enter a market, the larger the domestic sales of those firms.

5.6 Normalized export intensity

The paper defines normalized export intensity as

$$\frac{X_{nF}(j)/\bar{X}_{nF}}{X_{FF}(j)/\bar{X}_{FF}}. \quad (9)$$

Interpretation: This ratio compares a firm's relative foreign sales to its relative domestic sales. It isolates how important the foreign market is for the firm after controlling for firm size.

6 Estimation and identification

6.1 Parameterization

The model is estimated with five main parameters:

$$\Theta = (\theta, \lambda, \sigma_a, \sigma_\eta, \rho), \quad (10)$$

where θ controls efficiency dispersion, λ controls buyer reach or market penetration, σ_a controls demand-shock dispersion, σ_η controls entry-shock dispersion, and ρ is the correlation between demand and entry shocks.

Interpretation: These parameters determine how much observed entry and sales variation comes from common firm efficiency versus idiosyncratic market shocks.

6.2 Method of simulated moments

The authors simulate artificial French exporters for each candidate parameter vector. A simulated firm draws:

- a latent efficiency rank $u(j)$;
- destination-specific demand shocks $a_n(j)$;
- destination-specific entry shocks $\eta_n(j)$.

The model then determines:

- the destinations entered by each artificial firm;
- sales in each entered destination;
- domestic sales in France;
- export intensity.

The parameters are chosen so that simulated moments match the empirical moments in Figures 1–4 and Tables I–II.

Interpretation: The estimation uses the full pattern of where firms sell and how much they sell, rather than only aggregate trade flows.

6.3 Identification intuition

- The hierarchy of market entry identifies the importance of common firm efficiency.
- The similarity of sales distributions across markets identifies the common sales structure and shock dispersion.
- The relationship between French domestic sales and destination popularity identifies the efficiency dispersion parameter.

- Export intensity identifies the buyer-reach and market-penetration components.
- Deviations from strict hierarchy identify destination-specific entry and demand shocks.

Interpretation: Identification comes from requiring the same parameters to explain several distinct empirical regularities at once.

6.4 Economic interpretation of the estimates

- Underlying efficiency explains much of the extensive margin: which markets a firm enters.
- Conditional on entry, destination-specific shocks explain much of the intensive margin: how much the firm sells in that market.
- Entry shocks and demand shocks are essential because firms do not obey a perfect productivity hierarchy across all destinations.

Interpretation: The results split firm heterogeneity into a common component and market-specific components. This is the main quantitative anatomy of trade in the paper.

7 Counterfactual: a 10 percent decline in trade barriers

7.1 Aggregate effects

The paper simulates a 10 percent decline in bilateral trade barriers. The counterfactual produces:

- higher real wages in most markets;
- changes in relative wages across countries;
- more French exports to foreign markets;
- lower French domestic sales due to stronger import competition.

Interpretation: The counterfactual is general equilibrium: it changes both foreign opportunities and domestic competition.

7.2 Firm-level effects

- Total French firm sales rise by about \$16.4 billion.
- Export sales rise by about \$34.5 billion.
- Domestic sales fall by about \$18.1 billion.
- The number of exporting firms rises by about 10,716.
- The total number of French firms falls by about 26,589.

Interpretation: Trade liberalization does not simply expand all firms. It reallocates activity away from many lower-productivity firms toward top exporters.

7.3 Distributional effects across firm size

- Entry into exporting increases across many initial-size bins, but the largest export revenue gains are concentrated at the top.
- Nearly half the firms in the bottom decile disappear.

- Every lower decile has a decline in total sales.
- The top decile accounts for essentially all of the aggregate sales gain.
- Most of the export revenue increase comes from the top percentile.

Interpretation: The aggregate welfare or sales gain hides a sharp distributional reallocation. Export opportunities are not distributed evenly across firms.

8 Tables and figures

8.1 Figure 1: entry and sales by market size

Read:

- Panel A shows that the number of French firms selling in a market rises with market size.
- Panel B shows normalized entry: entry relative to French market share increases systematically with market size.
- Panel C shows that sales percentiles also rise with market size.

Economic message: Market size affects both the extensive margin of entry and the intensive margin of firm sales. This motivates a model in which larger markets attract more firms and allow larger sales scale.

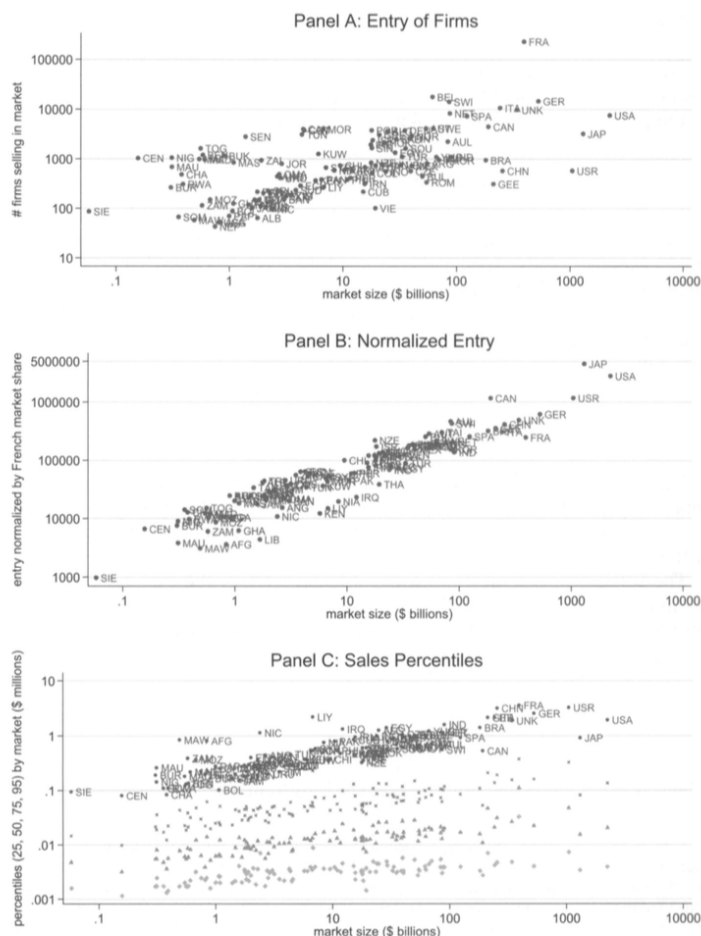


FIGURE 1.—Entry and sales by market size.

8.2 Tables I and II: top destinations and entry hierarchy

Read:

- Table I shows the seven most popular foreign destinations for French exporters.
- Belgium-Luxembourg is the most common foreign destination, with 17,699 French exporters.
- Table II shows that observed entry strings among top destinations are much more hierarchical than independent entry would predict.
- The model comes much closer to matching the observed hierarchy than the independence benchmark.

Economic message: Firms do not randomly choose export destinations. A common latent capability, interpreted as efficiency, is central to understanding market entry.

ANATOMY OF INTERNATIONAL TRADE
TABLE I
FRENCH FIRMS EXPORTING TO THE SEVEN MOST POPULAR DESTINATIONS

Export Destination	Number of Exporters	Fraction of Exporters
Belgium ^a (BE)	17,699	0.520
Germany (DE)	14,579	0.428
Switzerland (CH)	14,173	0.416
Italy (IT)	10,643	0.313
United Kingdom (UK)	9752	0.287
Netherlands (NL)	8294	0.244
United States (US)	7608	0.224
Any destination (all French exporters)	34,035	

^aBelgium includes Luxembourg.

Table I: top export destinations

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TABLE II
FRENCH FIRMS SELLING TO STRINGS OF TOP-SEVEN COUNTRIES

Export String ^a	Number of French Exporters		
	Data	Under Independence	Model
BE ^a	3988	1700	4417
BE-DE	863	1274	912
BE-DE-CH	579	909	402
BE-DE-CH-IT	330	414	275
BE-DE-CH-IT-UK	313	166	297
BE-DE-CH-IT-UK-NL	781	54	505
BE-DE-CH-IT-UK-NL-US	2406	15	2840
Total	9260	4532	9648

Table II: top-seven export strings

8.3 Figures 2 and 3: sales distributions and home-market size of exporters

Read:

- Figure 2 shows that sales distributions have similar shapes across Belgium-Luxembourg, France, Ireland, and the United States.
- Figure 3 shows that firms selling to more markets and to less popular destinations have higher domestic sales in France.
- The relationship between domestic sales and destination popularity is tight and approximately log-linear.

Economic message: The model must simultaneously explain similar sales distributions across markets and strong selection of larger domestic firms into harder export markets.



Figure 2: sales distributions by country

1

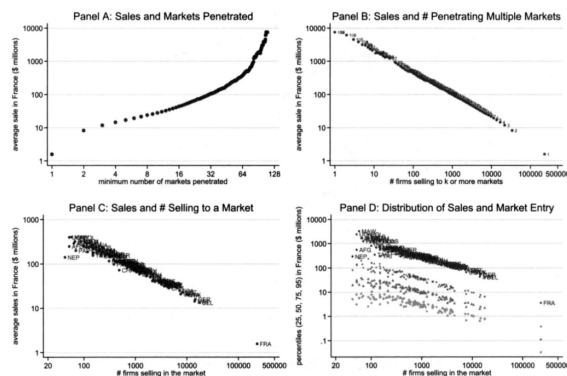


FIGURE 3.—Sales in France and market entry.

Figure 3: sales in France and market entry

8.4 Figures 4 and 5: export intensity and model fit

Read:

- Figure 4 shows that normalized export intensity rises with the number of French firms in a destination.
- Figure 5 compares model predictions to data for sales distributions, sales in France, and export intensity.
- The model captures the main slopes and distributional patterns, although it cannot reproduce every outlier.

Economic message: The estimated model fits the central empirical moments: entry, sales distribution shape, domestic sales selection, and export intensity.

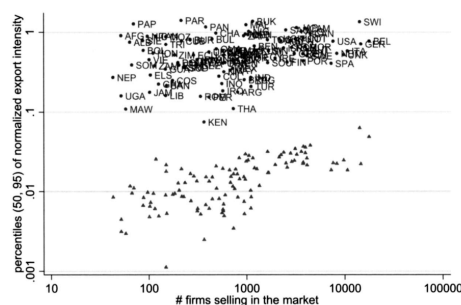


Figure 4: distribution of export intensity

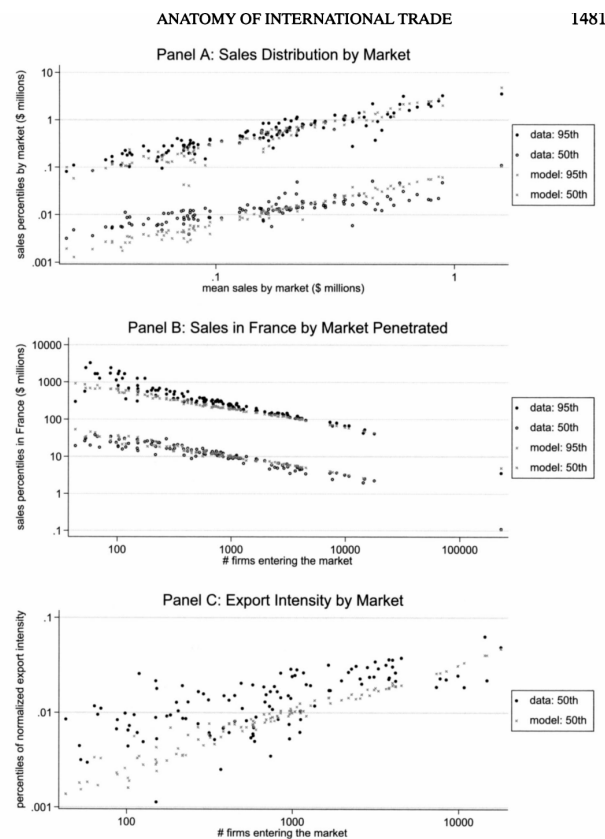


FIGURE 5.—Model versus data.

Figure 5: model versus data

8.5 Table III: aggregate outcomes of the trade-barrier counterfactual

Read:

- Table III reports country-level changes in real wages, relative wages, French firm sales, and numbers of French firms.
- French firm sales rise in many foreign markets, but the magnitude differs widely across destinations.
- Relative wages adjust, reflecting general-equilibrium effects of the trade-cost decline.

Economic message: The counterfactual is not a partial-equilibrium export shock. It changes wages, price indexes, market shares, and the number of French firms active in each market.

TABLE III
AGGREGATE OUTCOMES OF COUNTERFACTUAL EXPERIMENT

Country	Code	Counterfactual Changes (ratio of counterfactual to baseline)			
		Real Wage	Relative Wage	French Firms	
				Sales	Number
Afghanistan	AFG	1.01	0.92	1.22	1.23
Albania	ALB	1.01	0.94	1.35	1.34
Algeria	ALG	1.00	0.90	1.09	1.12
Angola	ANG	1.00	0.90	1.08	1.10
Argentina	ARG	1.01	0.96	1.57	1.52
Australia	AUL	1.02	0.96	1.35	1.29
Austria	AUT	1.04	1.04	1.49	1.32
Bangladesh	BAN	1.01	0.95	1.37	1.33
Belgium ^a	BEL	1.09	1.11	1.44	1.19
Benin	BEN	1.02	0.94	1.12	1.09
Bolivia	BOL	1.02	0.94	1.21	1.18
Brazil	BRA	1.01	0.96	1.64	1.57
Bulgaria	BUL	1.02	0.95	1.38	1.34
Burkina Faso	BUK	1.01	0.93	1.17	1.17
Burundi	BUR	1.01	0.92	1.21	1.21
Cameroon	CAM	1.01	0.92	1.19	1.19
Canada	CAN	1.04	1.05	1.43	1.26
Central African Republic	CEN	1.02	1.03	1.33	1.20
Chad	CHA	1.01	0.90	1.07	1.10
Chile	CHI	1.03	1.02	1.53	1.38
China	CHN	1.01	0.94	1.38	1.36
Colombia	COL	1.01	0.92	1.22	1.23
Costa Rica	COS	1.02	0.94	1.22	1.20
Cote d'Ivoire	COT	1.03	0.98	1.36	1.28
Cuba	CUB	1.01	0.93	1.26	1.24
Czechoslovakia	CZE	1.03	1.01	1.52	1.38
Denmark	DEN	1.04	1.06	1.46	1.27
Dominican Republic	DOM	1.04	0.99	1.36	1.28
Ecuador	ECU	1.02	0.96	1.33	1.28
Egypt	EGY	1.02	0.92	1.12	1.12
El Salvador	ELS	1.02	0.93	1.10	1.10
Ethiopia	ETH	1.01	0.92	1.08	1.09
Finland	FIN	1.03	1.02	1.53	1.38
France	FRA	1.02	1.00	0.95	0.88
Germany, East	GEE	1.01	0.96	1.58	1.52
Germany, West	GER	1.03	1.02	1.60	1.45
Ghana	GHA	1.02	0.99	1.38	1.28

Table III, part 1

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TABLE III—Continued

Country	Code	Counterfactual Changes (ratio of counterfactual to baseline)			
		Real Wage	Relative Wage	French Firms	
				Sales	Number
Papua New Guinea	PAP	1.07	1.09	1.33	1.13
Paraguay	PAR	1.01	0.93	1.21	1.20
Peru	PER	1.02	0.97	1.39	1.32
Philippines	PHI	1.02	0.97	1.50	1.43
Portugal	POR	1.03	1.03	1.47	1.32
Romania	ROM	1.01	0.97	1.68	1.61
Rwanda	RWA	1.00	0.90	1.14	1.17
Saudi Arabia	SAU	1.02	0.95	1.15	1.11
Senegal	SEN	1.03	1.01	1.36	1.24
Sierra Leone	SIE	1.03	1.17	1.36	1.08
Singapore	SIN	1.24	1.15	1.37	1.10
Somalia	SOM	1.03	0.96	1.09	1.05
South Africa	SOU	1.03	1.01	1.56	1.43
Spain	SPA	1.02	0.97	1.49	1.42
Sri Lanka	SRI	1.03	0.99	1.34	1.24
Sudan	SUD	1.00	0.91	1.13	1.15
Sweden	SWE	1.04	1.05	1.51	1.33
Switzerland	SWI	1.05	1.05	1.48	1.31
Syria	SYR	1.02	0.96	1.20	1.15
Taiwan	TAI	1.04	1.05	1.64	1.44
Tanzania	TAN	1.01	0.94	1.15	1.13
Thailand	THA	1.03	0.99	1.50	1.40
Togo	TOG	1.03	0.96	1.11	1.07
Trinidad and Tobago	TRI	1.04	1.01	1.22	1.12
Tunisia	TUN	1.04	1.00	1.36	1.26
Turkey	TUR	1.01	0.95	1.37	1.33
Uganda	UGA	1.00	0.90	1.06	1.08
United Kingdom	UNK	1.03	1.00	1.46	1.35
United States	USA	1.01	0.96	1.45	1.40
Uruguay	URU	1.02	1.00	1.65	1.53
USSR	USR	1.00	0.92	1.32	1.33
Venezuela	VEN	1.01	0.91	1.18	1.20
Vietnam	VIE	1.01	0.95	1.37	1.33
Yugoslavia	YUG	1.02	0.97	1.48	1.41
Zaire	ZAI	1.06	1.21	1.37	1.04
Zambia	ZAM	1.03	1.12	1.49	1.22
Zimbabwe	ZIM	1.02	0.97	1.43	1.36

TABLE III—Continued

Country	Code	Counterfactual Changes (ratio of counterfactual to baseline)			
		Real Wage	Relative Wage	French Firms	
				Sales	Number
Greece	GRE	1.02	0.97	1.37	1.30
Guatemala	GUA	1.01	0.92	1.16	1.17
Honduras	HON	1.02	0.95	1.19	1.16
Hong Kong	HOK	1.14	1.20	1.33	1.02
Hungary	HUN	1.05	1.04	1.41	1.25
India	IND	1.01	0.95	1.40	1.37
Honduras	HON	1.02	0.95	1.19	1.16
Hong Kong	HOK	1.14	1.20	1.33	1.02
Hungary	HUN	1.05	1.04	1.41	1.25
India	IND	1.01	0.95	1.40	1.37
Indonesia	INO	1.02	0.96	1.44	1.38
Iran	IRN	1.01	0.93	1.16	1.16
Iraq	IRQ	1.04	0.94	1.08	1.06
Ireland	IRE	1.07	1.09	1.43	1.21
Israel	ISR	1.04	1.01	1.44	1.31
Italy	ITA	1.02	0.99	1.57	1.46
Jamaica	JAM	1.05	1.02	1.35	1.22
Japan	JAP	1.01	0.98	1.80	1.69
Jordan	JOR	1.03	0.95	1.16	1.13
Kenya	KEN	1.01	0.93	1.18	1.18
Korea, South	KOR	1.04	1.04	1.58	1.40
Kuwait	KUW	1.02	0.94	1.14	1.12
Liberia	LIB	1.49	1.03	1.27	1.14
Libya	LIY	1.02	0.95	1.10	1.07
Madagascar	MAD	1.01	0.94	1.22	1.20
Malawi	MAW	1.01	0.92	1.17	1.17
Malaysia	MAY	1.07	1.08	1.45	1.23
Mali	MAL	1.02	0.95	1.15	1.12
Mauritania	MAU	1.08	1.19	1.36	1.05
Mauritius	MAS	1.07	1.05	1.47	1.29
Mexico	MEX	1.01	0.94	1.32	1.30
Morocco	MOR	1.02	0.98	1.37	1.30
Mozambique	MOZ	1.01	0.94	1.29	1.26
Nepal	NEP	1.01	1.01	1.35	1.24
Netherlands	NET	1.06	1.14	1.41	1.14
New Zealand	NZE	1.03	1.00	1.45	1.33
Nicaragua	NIC	1.01	0.89	1.06	1.09
Niger	NIG	1.02	1.09	1.47	1.25
Nigeria	NIA	1.00	0.89	1.07	1.12
Norway	NOR	1.04	1.03	1.38	1.23

Table III, part 2

8.6 Tables IV–VI: firm-level decomposition of the counterfactual

Read:

- Table IV shows that total French firm sales rise by \$16.4 billion, exports rise by \$34.5 billion, and domestic sales fall by \$18.1 billion.
- Table V shows that nearly half of firms in the bottom decile disappear, while many new exporters enter.
- Table VI shows that every lower decile loses total sales, while the top decile gains strongly.
- The top percentile accounts for most of the increase in export revenue.

Economic message: Trade liberalization raises aggregate export sales but redistributes activity sharply toward large, efficient firms.

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TABLE IV
COUNTERFACTUALS: FIRM TOTALS^a

	Counterfactual		
	Baseline	Change From Baseline	Percentage Change
Number			
All firms	231,402	−26,589	−11.5
Exporting	32,969	10,716	32.5
Values (\$ millions)			
Total sales	436,144	16,442	3.8
Domestic sales	362,386	−18,093	−5.0
Exports	73,758	34,534	46.8

Table IV: firm totals

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TABLE V
COUNTERFACTUALS: FIRM ENTRY AND EXIT BY INITIAL SIZE

Initial Size Interval (percentile)	All Firms			Exporters		
	Baseline No. of Firms	Counterfactual		Baseline No. of Firms	Counterfactual	
		Change From Baseline	Change (%)		Change From Baseline	Change (%)
Not active	0	1118	—	0	1118	—
0 to 10	23,140	−11,551	−49.9	767	15	2.0
10 to 20	23,140	−5702	−24.6	141	78	55.1
20 to 30	23,140	−3759	−16.2	181	192	106.1
30 to 40	23,140	−2486	−10.7	357	357	100.0
40 to 50	23,140	−1704	−7.4	742	614	82.8
50 to 60	23,138	−1141	−4.9	1392	904	65.0
60 to 70	23,142	−726	−3.1	2450	1343	54.8
70 to 80	23,140	−405	−1.8	4286	1829	42.7
80 to 90	23,140	−195	−0.8	7677	2290	29.8
90 to 99	20,826	−38	−0.2	12,807	1915	15.0
99 to 100	2314	0	0.0	2169	62	2.8
Total	231,402	−26,589		32,969	10,716	

Table V: entry and exit by initial size

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TABLE VI
COUNTERFACTUALS: FIRM GROWTH BY INITIAL SIZE

Initial Size Interval (percentile)	Total Sales			Exports		
	Baseline (\$ millions)	Counterfactual		Baseline (\$ millions)	Counterfactual	
		Change From Baseline	Change (%)		Change From Baseline	Change (%)
Not active	0	3	—	0	3	—
0 to 10	41	−24	−58.0	1	2	345.4
10 to 20	190	−91	−47.7	1	2	260.3
20 to 30	469	−183	−39.0	1	3	266.7
30 to 40	953	−308	−32.3	2	7	391.9
40 to 50	1793	−476	−26.6	6	18	307.8
50 to 60	3299	−712	−21.6	18	48	269.7
60 to 70	6188	−1043	−16.9	58	130	223.0
70 to 80	12,548	−1506	−12.0	206	391	189.5
80 to 90	31,268	−1951	−6.2	1085	1501	138.4
90 to 99	148,676	4029	2.7	16,080	11,943	74.3
99 to 100	230,718	18,703	8.1	56,301	20,486	36.4
Total	436,144	16,442		73,758	34,534	

9 Short summary

- The paper studies French manufacturing firms selling to 113 destinations.
- Entry into export markets increases systematically with market size.

- Entry into popular destinations follows a hierarchy, revealing a common firm component.
- Sales distributions are similar across destinations after normalization.
- Firms entering more and harder destinations have higher domestic sales.
- Export intensity is small relative to home sales but increases with destination popularity.
- The structural model includes efficiency, demand shocks, entry shocks, and fixed market-access costs.
- Simulated moments identify the importance of common efficiency versus destination-specific shocks.
- Efficiency explains much of market entry but less of conditional sales.
- A 10 percent trade-barrier reduction expands exports but hurts many smaller firms.
- Aggregate trade gains are concentrated among top firms and come with large reallocation.

10 Conclusion

10.1 Core conclusion

Eaton, Kortum, and Kramarz provide a structural anatomy of international trade using detailed firm-destination data. They show that a common efficiency component explains a large share of export participation, but destination-specific shocks are crucial for explaining sales conditional on entry.

10.2 Economic intuition

Efficient firms are larger at home and can overcome more foreign entry barriers, so they enter more and less popular destinations. Once a firm enters a destination, however, sales depend heavily on demand and entry shocks specific to that firm-destination pair. Therefore, exporting is strongly selected by efficiency, but conditional export sales are not simply a monotone transformation of firm productivity.

10.3 Incremental contribution takeaway

The paper’s incremental contribution is to unify four objects in one estimated trade model: firm entry across many destinations, sales distributions within destinations, aggregate bilateral trade shares, and firm-level counterfactual reallocation. It moves the heterogeneous-firm trade literature from stylized selection facts to a quantitative anatomy of which firms win and lose from lower trade barriers.

10.4 Policy takeaway

Trade liberalization can raise aggregate sales and exports while generating substantial displacement among smaller firms. The gains from lower trade barriers are concentrated among the most productive firms, so aggregate trade policy has distributional consequences across firms even within the same country and sector.